TOR GROUPS COMPUTE THE BASE CHANGE OF TOR UNDER FLATNESS AND COLIMIT HYPOTHESES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Functoriality of Tor in the pair of algebras and the resulting comparison maps. the author cites Algebra, Section algebra-section-functoriality-tor at proof lines 14993-14997. Produces the comparison map whose being an isomorphism is proved.
- Computation of Tor by tensor products against a free resolution, and flatness of the base change. used at proof lines 14999-15010. Gives the identification after the first base change.
- Lazard's theorem writing a flat module as a filtered colimit of finite free modules, and compatibility of Tor with filtered colimits. the author cites Algebra, Theorem algebra-theorem-lazard at proof lines 15012-15014. Upgrades the comparison map to an isomorphism.

Source reference. Original lemma 08HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10794–10794. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00M7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 813–813. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 058G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6779–6779. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Consider a commutative diagram of rings*

$$\begin{array}{ccccc} A' & \longleftarrow & R' & \longrightarrow & B' \\ \uparrow & & \uparrow & & \uparrow \\ A & \longleftarrow & R & \longrightarrow & B \end{array}$$

Assume that R' is flat over R and A' is flat over $A \otimes_R R'$ and B' is flat over $R' \otimes_R B$. Then

$$Tor_i^R(A, B) \otimes_{(A \otimes_R B)} (A' \otimes_{R'} B') = Tor_i^{R'}(A', B')$$

Proof. By Algebra, Section 00M7 there are canonical maps

$$\mathrm{Tor}_i^R(A, B) \longrightarrow \mathrm{Tor}_i^{R'}(A \otimes_R R', B \otimes_R R') \longrightarrow \mathrm{Tor}_i^{R'}(A', B')$$

These induce a map from left to right in the formula of the lemma.

Take a free resolution $F_\bullet \to A$ of A as an R-module. Then we see that $F_\bullet \otimes_R R'$ is a resolution of $A \otimes_R R'$. Hence $\mathrm{Tor}_i^{R'}(A \otimes_R R', B \otimes_R R')$ is computed by $F_\bullet \otimes_R B \otimes_R R'$. By our assumption that R' is flat over R, this computes $\mathrm{Tor}_i^R(A, B) \otimes_R R'$. Thus $\mathrm{Tor}_i^{R'}(A \otimes_R R', B \otimes_R R') = \mathrm{Tor}_i^R(A, B) \otimes_R R'$ (uses only flatness of R' over R).

By Lazard's theorem (Algebra, Theorem 058G) we can write A', resp. B' as a filtered colimit of finite free $A \otimes_R R'$, resp. $B \otimes_R R'$-modules. Say $A' = \operatorname{colim} M_i$ and $B' = \operatorname{colim} N_j$. The result above gives

$$\mathrm{Tor}_i^{R'}(M_i, N_j) = \mathrm{Tor}_i^R(A, B) \otimes_{A \otimes_R B} (M_i \otimes_{R'} N_j)$$

as one can see by writing everything out in terms of bases. Taking the colimit we get the result of the lemma. □